

CRISISWALLET(Java Project)

1. Abstract

CrisisWallet is a Java-based personal financial crisis impact simulator developed using **Java Swing**. The main purpose of the project is to help an individual understand how an economic crisis, supply disruption, war, pandemic, fuel shock, or other unexpected event can affect their personal monthly budget.

In normal circumstances, a person may have a fixed salary, monthly expenses, and some savings. However, during a crisis, the prices of essential goods and services may increase while income may remain the same or even decrease. This can significantly reduce the amount of money available for savings and can create financial stress.

CrisisWallet allows the user to enter their monthly salary, existing savings, and expenses in different categories such as **Food & Groceries, Fuel/Petrol, Transport, Rent/Housing, Utilities, Healthcare, Shopping, and Entertainment**. The user can then select a predefined crisis scenario or create a custom crisis by entering their own percentage changes.

The system calculates the effect of the crisis on the user's expenses, savings, personal inflation rate, additional monthly and yearly expenditure, emergency-fund coverage, and future savings. It also provides a **Budget Rescue Plan** and **Survival Mode** to make the results more practical and understandable.

The project additionally provides a comparison of different crisis scenarios and generates a final financial report that can be saved as a text file.

The application demonstrates several important Java programming concepts including **object-oriented programming, inheritance, abstraction, interfaces, polymorphism, arrays, collections, exception handling, file handling, multithreading, Swing GUI programming, event handling, graphics, and packages**.

2. Introduction

Financial planning is important for every individual because income and expenses are not always stable. Unexpected events can change the cost of living significantly.

For example, suppose a person earns **₹50,000 per month** and normally spends:

Expense Category	Monthly Amount (₹)
Food & Groceries	8,000
Fuel	4,000
Transport	2,000
Housing	12,000

Expense Category	Monthly Amount (₹)
Utilities	3,000
Healthcare	2,000
Shopping	3,000
Entertainment	2,000
Total Normal Expenses	36,000

Under normal conditions, the person may have a reasonable amount left for savings. Now suppose a crisis causes fuel, food, transportation, utility, and healthcare costs to increase, and at the same time the person's salary is reduced.

The question becomes:

"How much will this crisis actually affect my personal finances?"

CrisisWallet is designed to answer this question. Instead of simply showing general information about inflation or economic crises, the system takes an individual's own financial situation and calculates the **personalized impact**.

3. Problem Statement

During economic or social crises, individuals often face increases in the prices of essential goods and services. At the same time, salaries may remain unchanged, decrease, or become uncertain.

Most people may know that prices are increasing, but they may not know:

- How much extra they will spend every month.
- Which expense category affects them the most.
- How much their monthly savings will decrease.
- How much additional income they would need.
- How long their existing savings could support them.
- What their savings may look like after several months.
- Which expenses they should reduce first.
- Which crisis scenario would affect them the most.

Traditional budgeting applications generally focus on normal monthly budgeting. They do not necessarily provide a dedicated **crisis simulation and survival analysis**.

Therefore, there is a need for a system that can simulate different crisis situations and translate general price changes into an individual's personal financial consequences.

4. Proposed Solution

CrisisWallet provides a personalized financial crisis simulation system.

The user provides:

- Monthly salary
- Current savings
- Monthly expenses
- Selected crisis
- Optional salary reduction

The application then:

- Calculates normal monthly expenses.
- Applies crisis-specific percentage changes.
- Calculates crisis-level expenses.
- Calculates additional monthly and yearly expenses.
- Calculates personal inflation.
- Calculates normal and crisis savings.
- Calculates required additional income.
- Calculates emergency-fund coverage.
- Projects future savings.
- Provides a budget rescue plan and a survival-mode assessment.
- Compares different crisis scenarios.
- Generates a final report.

5. Objectives of the Project

The main objectives of CrisisWallet are:

- **5.1 Personalized Crisis Analysis** – To calculate how a particular crisis affects an individual's own financial situation.
- **5.2 Expense Impact Calculation** – To determine how much each expense category changes during a crisis.
- **5.3 Savings Analysis** – To determine how much monthly savings are reduced due to increased expenses and reduced income.
- **5.4 Emergency Fund Analysis** – To determine approximately how many months the user's existing savings could support their crisis-level expenses if income stopped.
- **5.5 Crisis Comparison** – To compare different crisis scenarios and determine which one has the greatest financial impact on the user.
- **5.6 Budget Rescue** – To provide a simple rule-based strategy for recovering lost savings by reducing selected discretionary expenses.

- **5.8 Survival Analysis** – To give the user a simple financial risk classification: Strong, Caution, or High Risk.
- **5.9 Report Generation** – To generate a complete financial crisis report that can be saved as a text file.
- **5.10 Demonstration of Java Concepts** – To implement important concepts from the Java syllabus in a practical real-world project.

6. Scope of the Project

The project focuses on **personal financial crisis simulation**. The system covers:

- Personal income
- Personal savings
- Household expenses
- Crisis-driven price changes
- Salary reduction
- Savings reduction
- Emergency fund
- Future savings projection
- Crisis comparison
- Budget reduction strategy
- Financial report generation

The system can be used as an educational and planning tool to understand the financial consequences of different crisis scenarios. However, the project does not attempt to predict actual future economic conditions. The crisis percentages used by the application are **scenario assumptions**.

7. Main Features

7.1 User Financial Setup

The user can enter the following details:

- Monthly salary
- Current savings
- Food & grocery expenses
- Fuel expenses
- Transport expenses
- Housing / rent
- Utilities
- Healthcare

- Shopping
- Entertainment

The application also provides a **Fill Sample Data** option to quickly test the system.

7.2 Crisis Selection

The application contains several predefined crisis scenarios:

Crisis Scenario	What It Represents
COVID-19 Pandemic	Effect of lockdowns, supply-chain disruption, and increased medical expenses.
Russia-Ukraine War	Effect of oil, gas, wheat, edible-oil, and other price shocks.
Middle East Conflict & Regional Instability	Potential effects of crude oil volatility and shipping disruptions.
LPG Cooking Gas Shortage	Increased utility-related expenses and shortages affecting household costs.
Custom Crisis	The user creates their own crisis by manually entering percentage changes for each expense category.

8. System Modules

The application is divided into several modules.

8.1 Setup Module

The Setup module collects the user's financial information: salary, savings, expenses, crisis, and salary cut. After entering the information, the user clicks **Simulate Crisis**. The system validates the data before starting the simulation.

8.2 Crisis Impact Module

This module calculates the direct financial impact of the selected crisis. For every category:

$$\text{Crisis Expense} = \text{Normal Expense} + (\text{Normal Expense} \times \text{Crisis Percentage} / 100)$$

Worked example for the food category:

Item	Value
Normal food expense	₹8,000
Crisis increase	10%
Calculation	$\text{₹8,000} + (\text{₹8,000} \times 10 / 100)$
Crisis expense	₹8,800
Additional expense	₹800

The system performs this calculation for every category.

9. Personal Inflation

The project calculates **personal inflation** based on the user's own spending pattern. The formula used is:

$$\text{Personal Inflation} = ((\text{Crisis Total Expense} - \text{Normal Total Expense}) / \text{Normal Total Expense}) \times 100$$

This is different from simply using a general inflation number because different individuals spend different amounts in different categories. For example, a person who spends a large amount on fuel will be more affected by a fuel crisis than someone who rarely uses a vehicle. Therefore, CrisisWallet provides a more personalized interpretation.

10. Savings Analysis

The Savings module determines how the crisis changes the user's ability to save.

Quantity	Formula
Normal Savings	Salary – Normal Expenses
Crisis Salary (if salary cut applied)	Salary × (1 – Salary Cut Percentage / 100)
Crisis Savings	Crisis Salary – Crisis Expenses
Savings Reduction	Normal Savings – Crisis Savings

Warning: If crisis savings become negative, the system warns the user that **"Expenses are higher than income."** This is one of the important practical outcomes of the project.

11. Extra Income Calculation

The system also calculates how much additional income is required to maintain the user's previous savings level.

Quantity	Formula
Required Salary	Crisis Expenses + Normal Savings
Extra Income Needed	Required Salary – Crisis Salary

If the result is negative, the system reports zero additional income needed. This helps answer: **"How much more should I earn every month to maintain my previous savings despite the crisis?"**

12. Emergency Fund Calculator

The Emergency Fund Calculator estimates how long the user's current savings could support their crisis-level expenses if their income completely stopped.

$$\text{Survival Months} = \text{Current Savings} / \text{Crisis Monthly Expenses}$$

Example Item	Value
Current savings	₹2,00,000
Crisis expenses	₹40,000 / month
Survival	$200,000 / 40,000 = 5$ months

The project presents this result in **Survival Mode**.

13. Survival Mode

Survival Mode answers a very practical question: **"If my income suddenly becomes zero, how long can I survive using my existing savings?"**

The system provides: savings available, crisis expenses per month, estimated survival duration, survival status, and progress visualization. The result is classified approximately as:

Status	Condition
Strong	Savings can cover 6 months or more
Caution	Savings can cover 3 to 6 months
High Risk	Savings cover less than 3 months

This gives the user a simple interpretation instead of only presenting a numerical result.

14. Projection Module

The Projection module estimates future savings over different time periods and compares savings **with crisis** against savings **without crisis**.

Feature	Details
Time periods supported	3 months, 6 months, 12 months, 24 months
Comparison	Savings with crisis vs. savings without crisis
Representation	Line chart

This allows the user to visually understand how a crisis can change their long-term financial position.

15. Budget Rescue Plan

One of the practical features of CrisisWallet is the **Budget Rescue Plan**. Instead of only informing the user that their savings have decreased, the system suggests where reductions can be made.

Category Type	Categories	Treatment
Discretionary	Entertainment Shopping Transport Food & Groceries	Targeted first for reduction
Essential	Rent / Housing Healthcare Utilities	Protected from being targeted first

The application uses predefined reduction rates to calculate a possible monthly recovery amount. The system then determines whether the lost savings can be fully recovered through these reductions.

Result	Meaning
Recoverable	The lost savings can be fully recovered through the suggested reductions.
Partial	Only part of the amount can be recovered; additional income or further expense reductions may be required.

16. Crisis Comparison Module

The Compare module allows the user to compare different crises. Each crisis is analyzed for:

- Additional monthly cost
- Additional yearly cost
- Personal inflation

The crises are then ranked according to financial impact. The system identifies:

- Hardest crisis for the user
- Mildest crisis for the user
- Number of crises compared

This is useful because a crisis that appears severe globally may not necessarily be the most damaging crisis for every individual.

17. Final Report Generation

The Report module generates a complete financial report that can also be saved as a **.txt** file. The report contains:

1. Crisis Impact

- Expense before crisis
- Expense during crisis
- Additional expense
- Personal inflation
- Biggest impact category

2. Savings Impact

- Salary before crisis
- Salary during crisis
- Normal savings
- Crisis savings
- Savings reduction
- Required salary
- Additional income needed

3. Emergency Fund

- Current savings
- Crisis expenses
- Emergency fund coverage

4. Budget Rescue Plan

- Amount to recover
- Suggested cuts
- Planned monthly reduction
- Rescue status

5. Survival Mode

- Savings available
- Crisis expenses
- Estimated survival
- Survival status

18. System Architecture

The project follows a modular object-oriented architecture. The main package is **crisisswallet**, containing the following sub-packages and classes:

Package	Classes / Interfaces	Responsibility
crisisswallet	Main	Application entry point
model	User InvalidInputException	Handles user data and custom exception
crisis	Crisis ScenarioCrisis CovidCrisis RussiaUkraineCrisis MiddleEastCrisis LpgShortageCrisis CustomCrisis	Handles crisis definitions
calculator	Calculator ImpactCalculator SavingsCalculator EmergencyFundCalculator ProjectionCalculator CrisisResult	Handles calculations
gui	MainFrame, SetupPanel, DashboardPanel ImpactPanel, SavingsPanel, ProjectionPanel ComparePanel, BudgetRescuePanel SurvivalModePanel, ReportPanel BarChartPanel, LineChartPanel	Handles the user interface
task	ProjectionAnimator CompareWorker AnimationListener	Handles background processing
report	ReportGenerator	Handles report creation
util	Displayable Reportable MoneyFormat	Provides reusable utilities

This separation makes the application easier to understand, maintain, and extend.

19. Java Concepts Used

The project demonstrates a large number of Java concepts. The table below lists each concept, where it is used in CrisisWallet, and the example or purpose.

No.	Java Concept	Where Used in the Project	Example / Purpose
A. Object-Oriented Programming			

No.	Java Concept	Where Used in the Project	Example / Purpose
1	Classes and Objects	Entire project – used to represent users, crises, calculators, GUI components, reports, and workers/threads	User user = new User(...); new MainFrame() new ImpactCalculator(...)
2	Encapsulation	User class – private data with public accessor methods	private double salary; private double savings; getSalary(), getSavings() Protects the internal state of the object
3	Inheritance	Crisis hierarchy and Calculator hierarchy	Crisis → ScenarioCrisis → CovidCrisis, RussiaUkraineCrisis, MiddleEastCrisis, LpgShortageCrisis Calculator → Impact / Savings / EmergencyFund / Projection Calculator (Multilevel inheritance)
4	Abstraction	Abstract classes Crisis and Calculator	public abstract Map<String, Double> getPercentages(); public abstract void calculate(); Every subclass must provide its own implementation
5	Polymorphism	Common Crisis reference for all crisis objects	Crisis chosen; Can refer to CovidCrisis, RussiaUkraineCrisis, MiddleEastCrisis, LpgShortageCrisis, CustomCrisis (Runtime polymorphism)
6	Method Overriding	Crisis, Calculator, and GUI classes	describe(), getPercentages(), calculate(), getSummary(), toReportText(), paintComponent()
7	Interfaces	Displayable, Reportable, AnimationListener	Calculator implements Displayable, Reportable (Multiple inheritance using interfaces)
8	Constructors	User class and crisis subclasses	public User() { this(0, 0); } super(...) to call the parent constructor
9	Static Members	Data and methods belonging to the class	public static final String[] CATEGORIES private static int userCount Static utility methods in MoneyFormat and ReportGenerator
10	final Keyword	Methods and classes that must not be changed	public final String getName() LpgShortageCrisis is a final class MoneyFormat and ReportGenerator are final
11	Object Class and toString()	User and Crisis classes	Overrides toString() to give meaningful textual representation of objects
12	Packages	model, crisis, calculator, gui, task, report, util	Improves organization, modularity, maintainability, readability, and separation of responsibilities
B. Arrays and Collections			

No.	Java Concept	Where Used in the Project	Example / Purpose
13	One-Dimensional Arrays	Expense categories and predefined crisis percentages	String[] CATEGORIES Percentage arrays for predefined crises
14	Two-Dimensional Arrays	ProjectionCalculator	double[][] table Stores Month, Savings with crisis, Savings without crisis
15	Collections Framework	Crisis lists, category maps, comparison results	See the Collections table below
16	Comparable Interface	CrisisResult in the Compare module	CrisisResult implements Comparable<CrisisResult> Sorts by additional monthly cost (largest first)
C. Exception Handling and File Handling			
17	Exception Handling	Input validation (checked and unchecked exceptions)	Custom checked exception: InvalidInputException Catches NumberFormatException and shows a meaningful validation message
18	throw and throws	Validation methods	throws InvalidInputException on methods that may fail throw to reject invalid values
19	finally Block	Report saving mechanism	File writer is closed whether or not an exception occurs (proper resource handling)
20	File Handling	Saving the final report as a text file	File, FileWriter, IOException
D. Swing GUI Programming			
21	Java Swing	Complete desktop GUI	JFrame, JPanel, JButton, JLabel, JTextField, JTextArea, JTable, JComboBox, JProgressBar, JScrollPane
22	Layout Managers	Dashboard cards, forms, statistics, comparison and projection controls	GridLayout (used extensively) BorderLayout (major interface sections)
23	Event Handling	Buttons and selection controls	ActionListener Simulate Crisis, select crisis, run comparison, animate projection, save report, navigate screens
24	CardLayout	MainFrame – switching between screens	Dashboard, Setup, Impact, Savings, Projection, Compare, Report, Budget Rescue, Survival Mode
25	Graphics	BarChartPanel and LineChartPanel	Graphics2D, paintComponent() Bar chart for expense impact; line chart for savings projection
E. Multithreading			

No.	Java Concept	Where Used in the Project	Example / Purpose
26	Multithreading	task package – ProjectionAnimator, CompareWorker	ProjectionAnimator implements Runnable (draws projection month by month) Compare module uses worker threads with Thread.start() and Thread.join()
27	Swing Thread Safety	Updating the GUI after background work	SwingUtilities.invokeLater() GUI updates run on the Event Dispatch Thread

Collections Used (Concept 15 in detail)

Collection Class	Used For
ArrayList	Storing lists of crises and worker objects
HashMap	Storing category-to-percentage relationships
LinkedHashMap	Maintaining expense information while preserving insertion order
LinkedHashSet	Storing categories that increased, without duplicates
Collections.sort()	Sorting crisis comparison results

Inheritance Hierarchies (Concept 3 in detail)

Hierarchy	Parent Class	Child Classes	Type
Crisis hierarchy	Crisis → ScenarioCrisis	CovidCrisis RussiaUkraineCrisis MiddleEastCrisis LpgShortageCrisis	Multilevel inheritance
Calculator hierarchy	Calculator	ImpactCalculator SavingsCalculator EmergencyFundCalculator ProjectionCalculator	Single inheritance

20. Data Validation

The system validates user input. This prevents incorrect input from producing meaningless calculations. Examples of invalid inputs include:

- Empty salary
- Negative salary
- Negative savings
- Negative normal expenses
- Invalid numbers (for example, text instead of a number)
- Salary cut greater than 100%
- Invalid custom crisis values

21. Algorithm / Working of the System

The overall process of the application is as follows:

START

1. Enter salary and savings
2. Enter monthly expenses
3. Select crisis
4. Enter salary cut
5. Validate input
6. Calculate normal expenses
7. Apply crisis percentages
8. Calculate crisis expenses
9. Calculate personal inflation
10. Calculate savings impact
11. Calculate emergency fund
12. Generate future projection
13. Generate budget rescue plan
14. Calculate survival mode
15. Compare crises
16. Generate final report

END

22. Example of System Working

Consider a user with **Salary = ₹50,000** and **Savings = ₹2,00,000**. The user enters monthly expenses, and the application first calculates the total normal expenses. Suppose the selected crisis increases:

Expense Category	Crisis Increase
Food	15%
Fuel	20%
Transport	12%
Housing	3%
Utilities	10%
Healthcare	5%
Shopping	6%
Entertainment	2%

The application applies each percentage to the corresponding personal expense. The difference between normal and crisis monthly expenses becomes the user's additional monthly financial burden. If the user's savings were ₹15,000 per month before the crisis and only ₹5,000 afterward:

Item	Value
Savings before crisis	₹15,000 per month
Savings after crisis	₹5,000 per month
Savings reduction	₹10,000 per month
Annual impact	$₹10,000 \times 12 = ₹1,20,000$ per year

This makes an abstract crisis much easier to understand from an individual's perspective.

23. Advantages of the Project

- **Personalized** – The calculations are based on the user's own expenses.
- **Easy to Understand** – The GUI presents complex calculations through cards, tables, charts, and simple messages.
- **Multiple Crisis Scenarios** – The user can compare different crises.
- **Custom Crisis** – Users can create their own scenarios.
- **Practical Financial Planning** – The project does not stop at showing the problem; it also provides a budget rescue strategy.
- **Emergency Fund Awareness** – Survival Mode helps users understand the importance of emergency savings.
- **Visual Representation** – Charts make financial changes easier to understand.
- **Educational** – The project demonstrates many important Java concepts in one application.

24. Limitations

Although the application is useful for educational simulation and personal planning, it has some limitations.

- **Scenario-Based Percentages** – The crisis percentages are assumptions and should not be considered exact historical or future economic data.
- **No Live Economic Data** – The application does not automatically retrieve live petrol, LPG, or food prices, inflation, gold prices, interest rates, or tax changes.
- **Simplified Expense Categories** – Real household spending can be much more complicated than the eight categories currently used.
- **Simplified Salary Model** – The system assumes a simple percentage-based salary cut.
- **No Investment Analysis** – Stocks, mutual funds, bonds, real estate, and other investments are not currently included.
- **No User Database** – The current version does not permanently store multiple user profiles in a database.

- **Rule-Based Budget Rescue** – Recommendations are based on predefined reduction priorities rather than advanced optimization or machine learning.

25. Future Enhancements

The project has significant potential for future development.

- **Live Economic Data** – Connect to APIs for real-time petrol, LPG, food prices, inflation, gold prices, and currency exchange rates to make the simulation more dynamic.
- **Database Integration** – Add a database such as MySQL to store user profiles, previous simulations, crisis scenarios, financial history, and saved reports.
- **Login System** – Add authentication so that each user has a private financial profile.
- **Advanced Budget Optimization** – Automatically determine the best combination of expense reductions, for example: "What is the minimum lifestyle reduction required to recover ₹10,000 per month?"
- **Investment Impact** – Include stocks, mutual funds, fixed deposits, gold, real estate, loans, and EMIs to simulate how a crisis affects both income and assets.
- **Loan and EMI Module** – Calculate EMI, interest, remaining loan balance, and the impact of interest-rate increases.
- **Family Budget Mode** – Support multiple household members: incomes, children's, household, education, and medical expenses.
- **Graphical Dashboard Improvements** – Pie charts, area charts, interactive graphs, expense heatmaps, financial risk gauges, and monthly trend graphs.
- **Mobile Application** – Convert the Java desktop application into a mobile or web-based application.
- **AI-Based Recommendations** – Analyze spending patterns and generate personalized suggestions.
- **Crisis Severity Levels** – Introduce Mild, Moderate, Severe, and Extreme crisis levels.
- **Historical Scenario Analysis** – Use verified historical data to ask: "What would my current budget have looked like during a particular historical crisis?"

26. Testing

The application should be tested using different categories of inputs.

Test Case	Scenario	Input	Expected Result
1	Normal Input	Salary = ₹50,000 Savings = ₹2,00,000 Valid expenses Crisis = COVID-19 Salary cut = 0%	Simulation completes successfully.
2	Negative Salary	Salary = -₹50,000	System rejects the input.

Test Case	Scenario	Input	Expected Result
3	Negative Expense	Food expense = -₹5,000	System displays an invalid-input warning.
4	Invalid Text	Salary = "abc"	Application catches the invalid number and displays an appropriate message.
5	Salary Cut > 100%	Salary cut = 150%	System rejects the value.
6	Custom Crisis	Custom percentages for all categories	Application calculates the crisis impact using those percentages.
7	Zero Expenses	Normal expenses are zero	Personal inflation does not cause a division-by-zero error.

27. Non-Functional Requirements

- **Performance** – Calculations should be completed quickly for normal user input.
- **Usability** – The interface should be easy for a non-technical user to understand.
- **Reliability** – Invalid input should not crash the application.
- **Maintainability** – The modular package structure allows individual components to be modified independently.
- **Portability** – The application can run on systems supporting Java.
- **Scalability** – New crisis classes and calculators can be added without redesigning the entire system.

28. System Requirements

28.1 Software Requirements

- Java Development Kit (JDK)
- Java compiler (compile with: javac)
- Java Swing libraries
- Operating system supporting Java
- Run the application with: java

28.2 Hardware Requirements

Basic hardware is sufficient. No specialized hardware is required. Recommended:

- Processor: Dual-core or better
- RAM: 4 GB or more
- Storage: Approximately 100 MB
- Display: Standard desktop / laptop display
- Keyboard and mouse

29. Design Principles Used

The project follows several good software design principles.

29.1 Separation of Responsibilities

Different packages handle different tasks:

- model handles user data.
- crisis handles crisis definitions.
- calculator handles calculations.
- gui handles the interface.
- report handles reports.
- task handles background processing.
- util provides reusable utilities.

29.2 Reusability

Classes such as StatCard and MoneyFormat can be reused throughout the application.

29.3 Extensibility

A new crisis can be created by extending the existing crisis hierarchy (Crisis → NewCrisis) without rewriting the entire calculation system.

30. Why Object-Oriented Programming Is Suitable

Object-oriented programming is particularly suitable for this project because the application contains many real-world entities:

- **User** represents a real person.
- **Crisis** represents an economic event.
- **Calculator** represents a financial calculation.
- **ReportGenerator** represents report creation.
- **CrisisResult** represents a crisis comparison result.

This allows the real-world problem to be mapped naturally into software objects.

31. Educational Significance

CrisisWallet is not only a financial application but also a demonstration of Java programming concepts. A single project demonstrates:

- Classes, Objects, and Constructors
- Encapsulation
- Inheritance and Multilevel inheritance
- Abstraction

- Polymorphism and Method overriding
- Interfaces
- Static and Final members
- Arrays and Two-dimensional arrays
- Collections and Comparable
- Exception handling
- File handling
- Packages
- Swing, Event handling, and Layout managers
- Graphics
- Multithreading

Therefore, the project provides a practical implementation of theoretical Java concepts.

32. Conclusion

CrisisWallet is a Java-based personal financial crisis simulation system designed to help users understand how unexpected crises can affect their financial stability. The application converts crisis assumptions into personalized financial consequences by analyzing the user's salary, savings, and expense structure.

It calculates:

- Crisis expenses
- Personal inflation
- Additional monthly cost
- Additional yearly cost
- Savings reduction
- Required income
- Emergency fund coverage
- Future savings
- Budget recovery
- Survival duration
- Crisis rankings

The addition of **Budget Rescue** and **Survival Mode** makes the project more practical because the system does not merely show that a financial problem exists; it also provides a simple interpretation of what the user can do about it.

From a programming perspective, the project successfully demonstrates a wide range of Java concepts including **OOP, inheritance, abstraction, polymorphism, interfaces, collections,**

exception handling, file handling, Swing GUI development, graphics, event handling, and multithreading.

The project can be further enhanced by integrating real-time economic data, databases, investment analysis, loan management, AI-based recommendations, and mobile/web support.

Overall, CrisisWallet demonstrates how Java programming can be used to solve a relatable real-world problem and transform complex financial concepts into an understandable interactive application.

33. Project Highlight / Unique Feature

A major strength of this project is that it is **not simply a budget calculator**. The core idea can be summarized as:

"What happens to MY wallet when a crisis happens?"

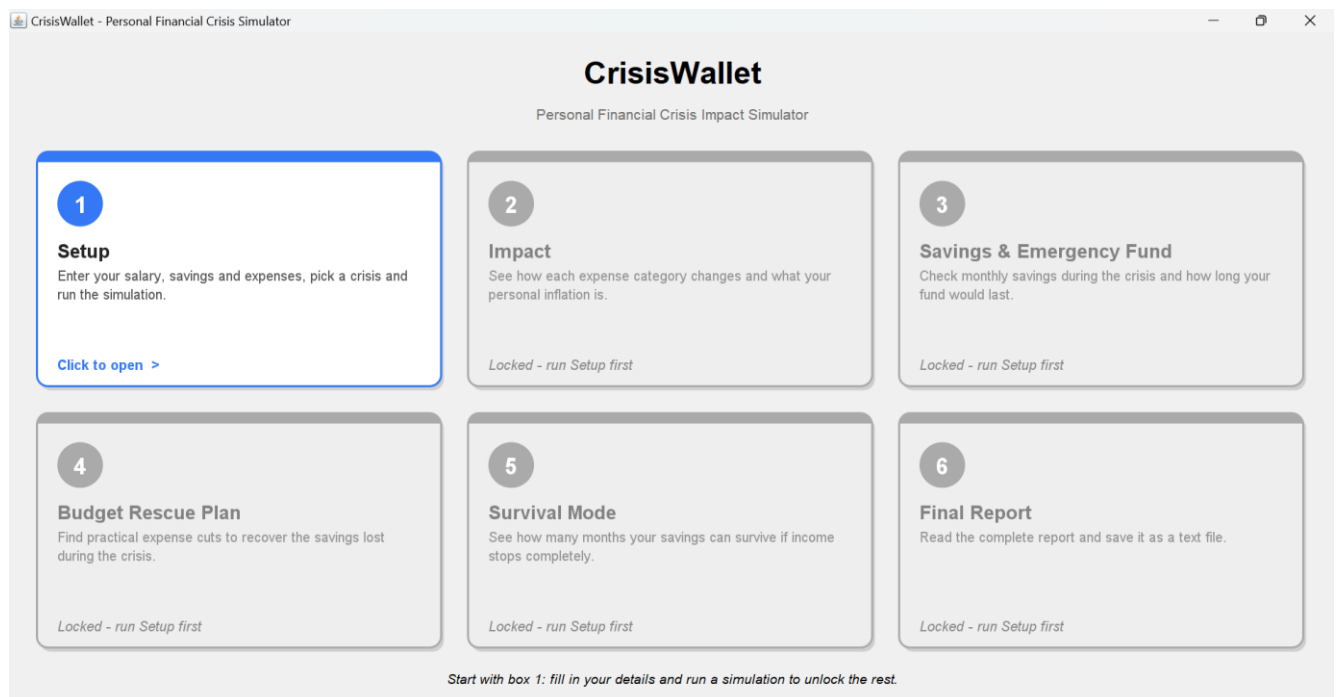
Instead of only saying: "Fuel prices increased by 20%."

CrisisWallet asks: **"You spend ₹4,000 on fuel. If fuel increases by 20%, how much extra will YOU spend, how much will YOUR savings fall, how long can YOUR savings survive, and what can YOU cut to recover the difference?"**

That personalized approach is the central concept of the project.

GITHUB LINK: [Bhavayami/CrisisWallet: Personal Budget & Crisis Impact Simulator](https://github.com/Bhavayami/CrisisWallet: Personal Budget & Crisis Impact Simulator)

Screenshots:



CrisisWallet - Personal Financial Crisis Simulator
1. Setup

1. Your financial profile

Monthly salary (Rs.)
Current savings (Rs.)
Monthly Food & Groceries
Monthly Fuel / Petrol
Monthly Transport
Monthly Rent / Housing
Monthly Utilities
Monthly Healthcare
Monthly Shopping
Monthly Entertainment

50000
200000
8000
4000
2000
12000
3000
2000
3000
2000

2. Choose a crisis

COVID-19 Pandemic
COVID-19 Pandemic
Russia-Ukraine War
Middle East Conflict & Regional Instability
LPG Cooking Gas Shortage
Custom Crisis (enter your own %)

Fuel / Petrol: +3%
Transport: +5%
Rent / Housing: +2%
Utilities: +5%

Custom crisis: % change per category

Food & Groceries
Fuel / Petrol
Transport
Rent / Housing
Utilities
Healthcare
Shopping
Entertainment

0
0
0
0
0
0
0
0

Salary cut during crisis (%)
5

Fill sample data
Simulate crisis

Simulated: COVID-19 Pandemic (2020)

CrisisWallet - Personal Financial Crisis Simulator
1. Setup

1. Your financial profile

Monthly salary (Rs.)
Current savings (Rs.)
Monthly Food & Groceries
Monthly Fuel / Petrol
Monthly Transport
Monthly Rent / Housing
Monthly Utilities
Monthly Healthcare
Monthly Shopping
Monthly Entertainment

50000
200000
8000
4000
2000
12000
3000
2000
3000
2000

2. Choose a crisis

Russia-Ukraine War
Russia-Ukraine War: Oil, gas, wheat and edible oil price shock (2022)

Scenario assumptions (% change):
Food & Groceries: +15%
Fuel / Petrol: +20%
Transport: +12%
Rent / Housing: +3%
Utilities: +10%

Custom crisis: % change per category

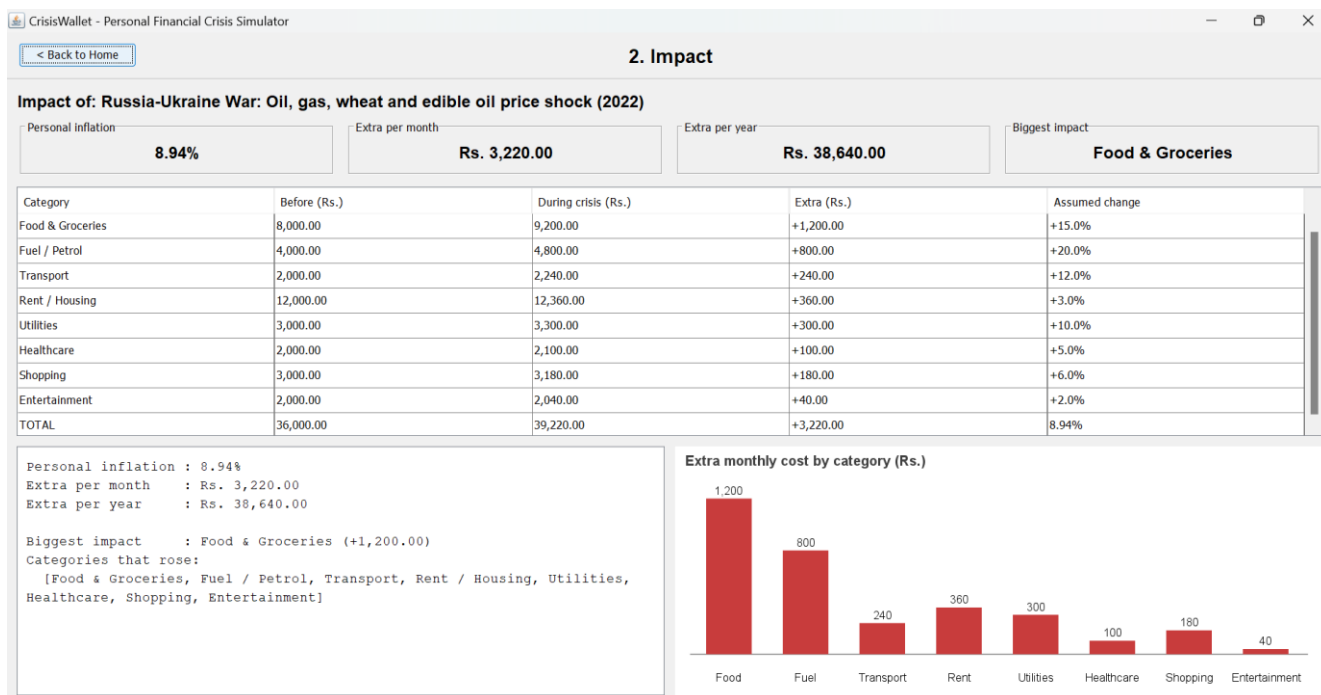
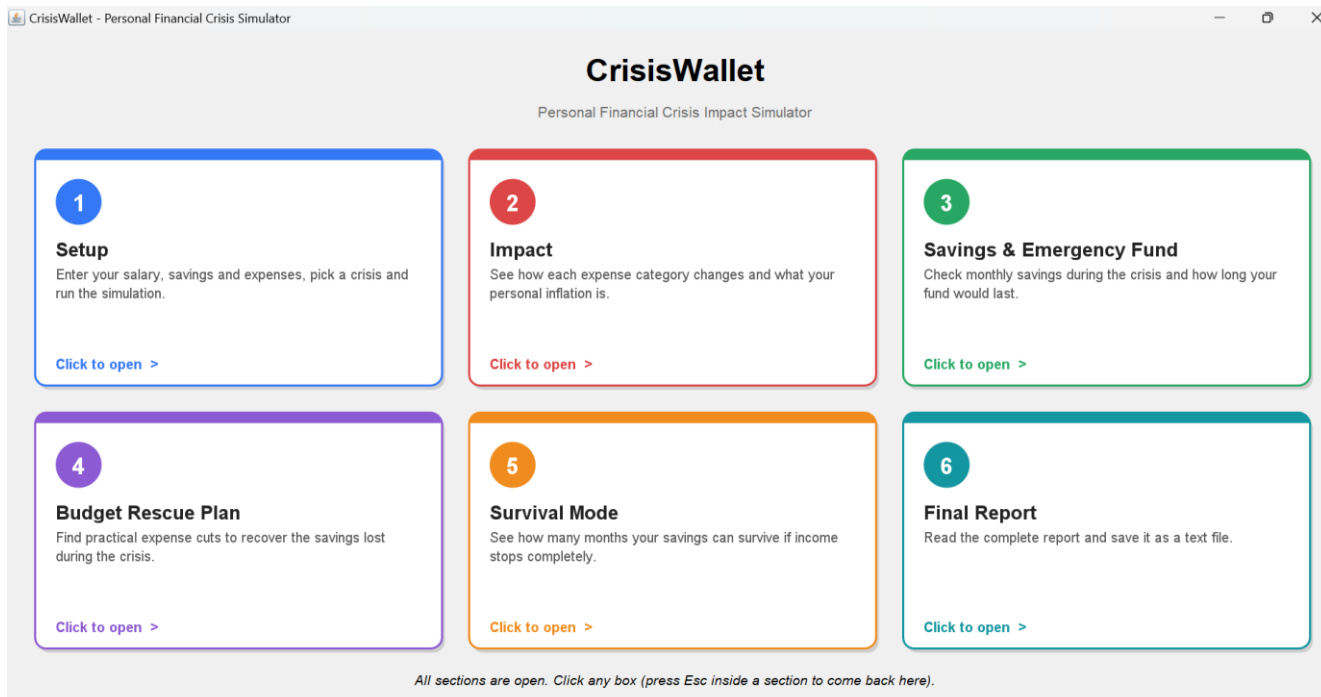
Food & Groceries
Fuel / Petrol
Transport
Rent / Housing
Utilities
Healthcare
Shopping
Entertainment

0
0
0
0
0
0
0
0

Salary cut during crisis (%)
5

Fill sample data
Simulate crisis

Simulated: COVID-19 Pandemic (2020)



CrisisWallet - Personal Financial Crisis Simulator

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3. Savings & Emergency Fund

Monthly savings before
Rs. 14,000.00

Monthly savings during
Rs. 8,280.00

Savings reduction
Rs. 5,720.00

Extra income needed
Rs. 5,720.00

Emergency fund lasts
5.1 months

MONTHLY SAVINGS

Salary before : Rs. 50,000.00

Salary during crisis : Rs. 47,500.00 (cut 5.0%)

Monthly savings before : Rs. 14,000.00 (28.00% of salary)

Monthly savings during : Rs. 8,280.00 (17.43% of salary)

Savings reduction : Rs. 5,720.00

Salary needed to keep old savings : Rs. 53,220.00

Extra income needed per month : Rs. 5,720.00

EMERGENCY FUND

Current savings : Rs. 200,000.00

Crisis expenses : Rs. 39,220.00 per month

Emergency fund coverage : 5.10 months (if income stops)

Emergency fund coverage

If your income stopped, how long would savings last?

5

10 months covered (full bar = 12 months)

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4. Budget Rescue Plan

Amount to recover
Rs. 5,720.00

Planned monthly cut
Rs. 3,389.00

Rescue status
Partial

BUDGET RESCUE PLAN

=====

Your crisis has changed your monthly savings by Rs. 5,720.00.

The plan below uses simple rules: discretionary spending is considered first, while essential expenses such as rent and healthcare are protected.

- Reduce Entertainment by Rs. 1,020.00 per month.
- Reduce Shopping by Rs. 1,113.00 per month.
- Reduce Transport by Rs. 336.00 per month.
- Reduce Food & Groceries by Rs. 920.00 per month.

This is a partial rescue plan. You would still need Rs. 2,331.00 more each month through extra income or further reductions.

Protected essentials

Rent / Housing, Healthcare and Utilities are not targeted first because they are essential household costs.

Crisis impact

Extra monthly cost : Rs. 3,220.00

Normal savings : Rs. 14,000.00

Crisis savings : Rs. 8,280.00

Salary needed to keep old savings : Rs. 53,220.00

Tip: Your wallet does not need panic first. It needs priorities. ☺

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5. Survival Mode

Months you can survive

5.10 months

Crisis expense / month

Rs. 39,220.00

Savings available

Rs. 200,000.00

Survival status

Caution

FINANCIAL SURVIVAL MODE
=====

Question: If your income stopped completely today, how long could your current savings cover your crisis-level expenses?

Current savings : Rs. 200,000.00
Crisis expenses : Rs. 39,220.00 per month
Estimated survival : 5.10 months
Survival status : Caution

You have a useful cushion, but a prolonged crisis could put pressure on your savings. Look for ways to reduce discretionary spending.

Monthly savings during the crisis: Rs. 8,280.00
Extra income needed to keep old savings: Rs. 5,720.00

Remember: this is a planning estimate. It assumes the crisis-level monthly expense stays constant and income is zero during the survival calculation.

Emergency survival meter

5.10 months covered (full bar = 12 months)

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6. Final Report

Save report as .txt

Copy to clipboard

CRISIS WALLET - FINAL REPORT
=====

Profile : Salary Rs. 50,000.00, savings Rs. 200,000.00, expenses Rs. 36,000.00
Crisis : Russia-Ukraine War: Oil, gas, wheat and edible oil price shock (2022)
Salary cut during crisis: 5.0%
(Percentages are scenario assumptions, not exact historical figures.)

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1. CRISIS IMPACT

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Category	Before	Crisis	Extra
Food & Groceries	8,000.00	9,200.00	+1,200.00
Fuel / Petrol	4,000.00	4,800.00	+800.00
Transport	2,000.00	2,240.00	+240.00
Rent / Housing	12,000.00	12,360.00	+360.00
Utilities	3,000.00	3,300.00	+300.00
Healthcare	2,000.00	2,100.00	+100.00
Shopping	3,000.00	3,180.00	+180.00
Entertainment	2,000.00	2,040.00	+40.00
TOTAL	36,000.00	39,220.00	+3,220.00

Personal inflation rate : 8.94%
Extra cost per month : Rs. 3,220.00
Extra cost per year : Rs. 38,640.00
Biggest impact : Food & Groceries (+1,200.00)
Categories that rose : [Food & Groceries, Fuel / Petrol, Transport, Rent / Housing, Utilities, Healthcare, Shopping, Entertainment]

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2. SAVINGS IMPACT